

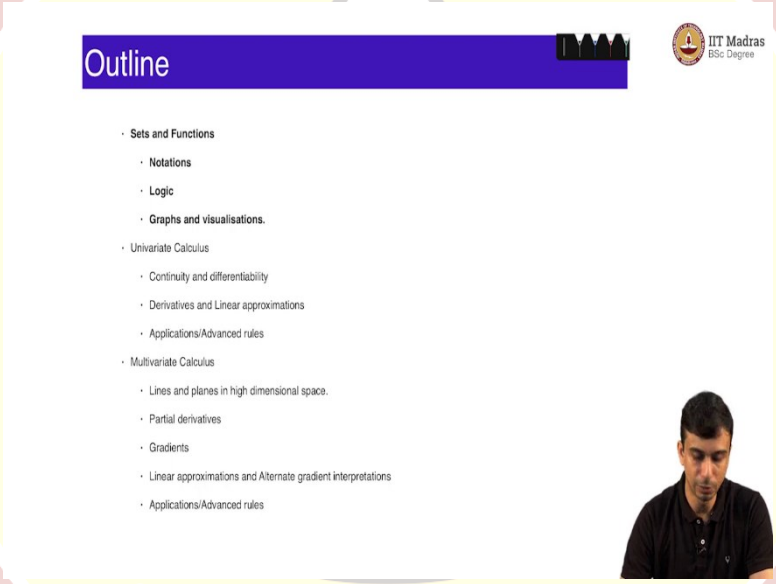
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Sets and Functions

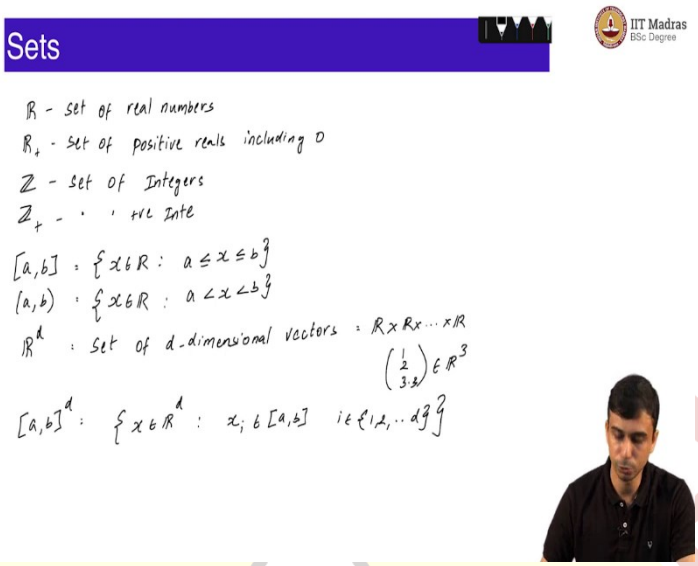
Hello everyone, welcome to the second week of the machine learning foundations course. In this week, we will look at the basic math tools that will be required for handling the rest of the course, I have titled it calculus because that is the majority of this course. And this will majorly overlap with the later part of the mathematics 2 course that you would have already done. But still, this has a different focus into some more geometric or a graphical focus rather than the course that you would have seen in math 2 for data science.

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So here is a brief overview of this week's content. We will first take a look at some basic sets and functions that we will be using throughout the course and notations and the notations for them. And some basic logic statements that we will be making, and their math notation for that. And we will wrap the first part of this week with how to visualize functions and there are several ways of visualizing these functions, how to graphically visualize these functions, that will be the final part of the first lecture today.

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Here are some basic sets that you will be using over and over throughout the course. The first set is R, which is simply the set of set of real numbers, which you would have seen already. And then you have R +, which is simply the set of real numbers, including 0 positive real numbers, including 0. Then we have Z, which is simply the set of integers. And Z + is a set of positive integers, including 0.

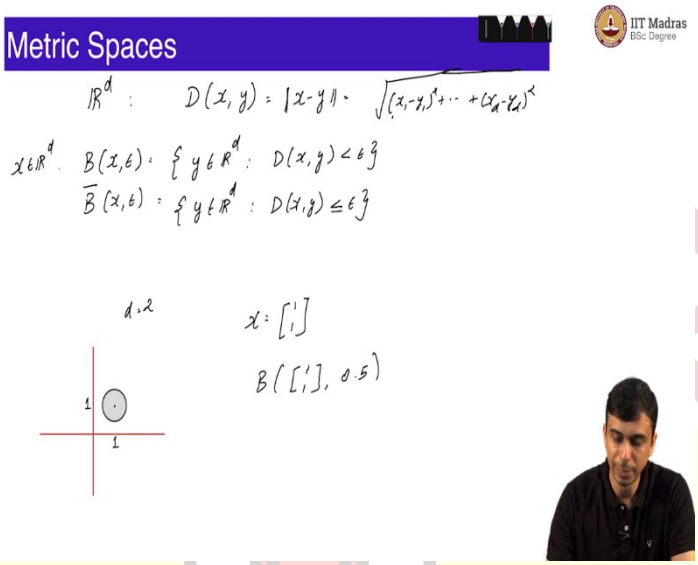
So, these are the four domain sets that will very often be using the course. And then there are a couple of other sets that we will use often, which is the closed interval a, b, this is the set of real numbers real numbers, such that they are between a and b, and the boundaries are inclusive, we include both a and be, so that is the closed interval. Then we have the open interval a , b, which is simply the set of all x in real a < x < b.

The only difference between these 2 sets is that in the second set, the boundary points a and b are not included. So, all of these sets can be exponentiated to power d, the Cartesian product d × will be taken for example, we can have Rd is simply the set of D dimensional vectors.

So, Rd is the same as writing R cross R cross d × an example element in for example, R3 would be 1 , 2 , 3.3 is an element in R3. You can have this power Cartesian product power to pretty much any set that you can think of for example, you can have a , b power d is simply the set of d dimensional vectors, but each element in the d dimensional vector is going to be between a and b.

So, we can write this compactly as the set of D dimensional vectors in Rd such that xi belong belongs to the closed interval a , b for all i belongs to 1 , 2 ,. So, these are some of the sets, the basic sets that we will notate throughout the course.

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The next object that we will be interested in, in analysing is a metric space. So, what is a metric space? A metric space is simply a set with an extra structure associated with it. And the means that structure is the distance function. So, you can have a distance function D, which takes in 2 objects the set that is the distance function.

So, that the default metric space that you will be working throughout is Rd which is the D dimensional vectors with the distance function simply given by the Euclidean distance by that I mean D of X , Y is simply the norm of x - y, which is simply the square root of x1 - y1, the whole squared + … xd - yd, the whole square. So, this is the standard Euclidean norm. This will be the default metric space for most of our course.

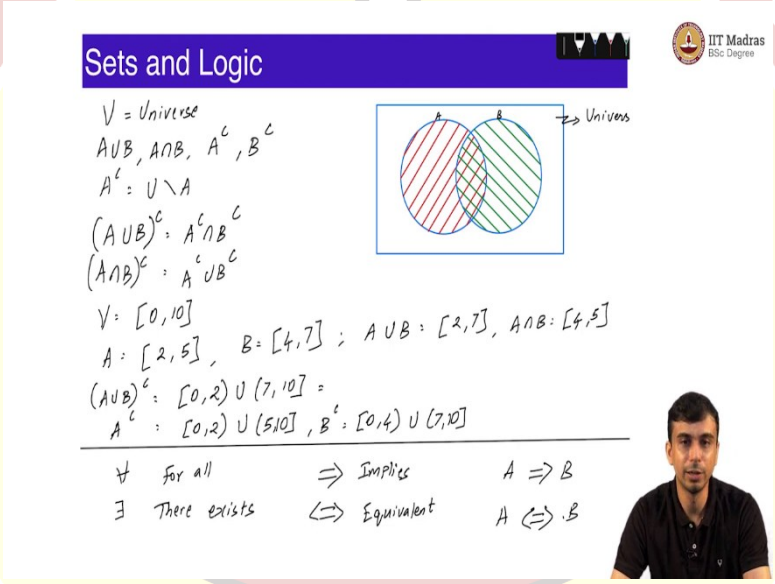
This metric, the there is a crucial set that is associated with metric space, which is a ball, we can define an open ball of radius epsilon centered at x, that is given by b of x , epsilon is simply the set of y in Rd, such that the distance of y from x is strictly < epsilon. This is the open ball where we have some x is some element in Rd. And ball of x , epsilon is simply the ball of radius epsilon centered at x.

Some× we will also be used the closed ball we also using the closed ball, which is B bar of x , epsilon, which is the set of y in Rd such that d of x , y < or =epsilon, the only difference between

the open ball and the closed ball is that the open ball does not contain the boundary, the closed ball does contain the boundary. What do I mean by that? Here is a simple example for the set.

So, imagine you are in D dimension, d =2 space, 2 dimensional space. And let us say for example, x is the vector 1 , 1. So here, this grey set is the ball of Centered at 1 , 1 and radius, let us say 0.5. This is the open ball, if you do not include the black boundary under it is actually =the close ball, if you include the boundary. If you do not include the boundary, it is the open ball, if you include the boundary, it is the closed ball. This is an example of a ball of radius or a given radius around a given center. The ball will be a very crucial part for doing a lot of analysis with metric spaces.

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So now, we move to another crucial aspect of the math that we will do here, which is some sets and logic. So, you might have seen Venn diagrams, in your school itself, but we will still do a quick recap of that. So, you have sets living in a universe. So, that is that universe, let us denote it by the set V, V is the universe. And you have, let us say, sets A and B, which contain elements in the universe.

The basic operations that are associated with sets are union and intersection. So, you can have, we use the universe. You can have A union B, A intersection B, and A complement, and B complement. So, these are some of the basic sets associated with a given that offsets the A and B, A union B is simply the union of both the A and B.

For example, if this was a, and this was B, union B is simply the entire all elements in A and P, A intersection B is the elements that are in both A and B. A complement is simply the set of

elements in the universe, but not in a and similarly B complement. So, you can in principle, write A compliment, as the universe set difference A. So, this is another notation that you might want to know this is the notion for set difference.

So, how do you denote the difference between 2 sets? This is the backslash symbol. That is A complement. Similarly, you can have B complement also, with the Venn diagram, you might have seen the De Morgan's laws, which are the basic laws of logic, essentially, what does it say? Well, it is a collection of 2 equations, which is A union B, the whole complement is simply =A compliment, intersection B complement.

And similarly, A intersection B, the whole complement is =A complement, union B complement. Graphically it is pretty easy to understand A union B the whole complement is simply the elements that are in the universe, but not neither A nor in B. So, that is essentially the unshaded region of this diagram, forms A union B the whole complement, that is clearly exactly =A complement intersection B complement.

So, that is the first law and similarly for the second law, A intersection B the whole complement is simply the set except for this intersection. That is A intersection B, the whole complement, and that is clearly exactly =A complement, union B complement, we will see this with a very simple example.

Let us say your universe V is the closed interval 0 , 10. This is the universe here. Let us say we have 2 sets A, A is =the closed interval, let us say 2 , 5 and let us say B is the closed interval given by 4 , 7. So, we have VA s, the closed interval 2 , 5, and B is the closed interval 4 , 7. Now let us just quickly verify the De Morgan's here.

What is A union B with this, we can say what is A union B union B is 2 , the interval 2 , 5 union the interval 4 , 7, which is clearly the interval 2 , 7. What is the A intersection B? It is clearly the interval, closed interval 4 , 5. So, what is A union B, the whole complement A union B the whole complement is simply the set 0 , the closed interval 0 10, difference, the set 2 7, and that is simply given by the union of the set 0 , 2, union 7 , 10.

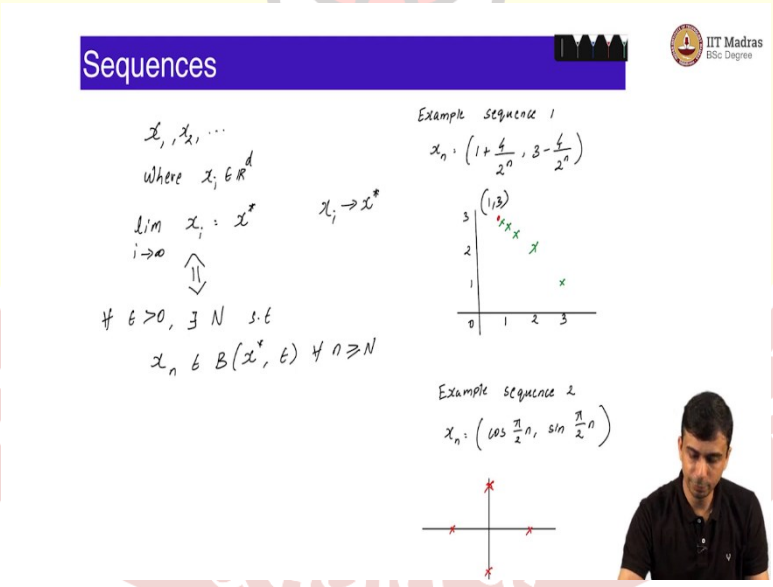
Note that I am using closed square brackets for 0 and 10. But parenthesis for 2 and 7 because 2 and 7 are not included in A union B, the whole complement. So, you can place A union B the whole complement is 0 , 2 union 7 , 10, with parenthesis for 2 and 7. This is clearly =A compliment, intersection B compliment, why is that A compliment is simply the we will just verify that quickly.

What is A compliment, well A compliment is the set 0 , 2 union 5 , 10 and B complement is simply the set 0 , 4 union 7 , the intersection of A complement and B complement is exactly =A union B the whole complement. So, this is a pretty simple way to simple verification, you can do a similar verification for A intersection B the whole complement also.

So, this is a DeMorgan's law and the basic set operations that you will need to be familiar and comfortable with. So, there is some logic statements that we will also be using throughout the course. And the four main types of logic modifiers that you will see are these first is you have inverted A, which is said to mean for all and you have an inverted E, which means there exists and then you have the arrow marks.

This, the double-sided arrow mark essentially means implies. And you might see a two-sided arrow mark which implies which saying equivalent to, so you can say A arrow B, A and B are two statements, you can say A arrow B is =saying that A implies B, you can have two sided arrows, which means that A is =B, A implies B and B implies A, then we have a two-sided arrow.

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So, we will illustrate all these logic statements through another building block of analysis and calculus, which is sequences. What is a sequence a sequence is simply an ordered collection of elements. That is a sequence. Here is an example sequence, here is a collection of elements in each of these elements are going to lie in R2 xn is 1 + 4 by 2 power n , 3 - 4 by 2 power n.

First, you can see that each of these elements are two dimensional. So, these elements are in R2. What for example, let us say set n =1 is that n =1 x1 is simply 1 + 2, , 3 - 2, which is 3 , 1

which is this element here. You can substitute n =2, you would get 1 + 2 ,1 + 1, 3 - 1, which is 2 , 2. And you can keep on increasing n and let us say for large enough n, we can you have 1 + 4 by 2 power n is approximately going to be =1, and 3 - 4 by 2 power n is approximately going to be =3.

So, if you this is the first element is x1, the second element is x2, and you can keep on doing this as your n increases, you can see that the xn seems to approach this point, the right point here, which is simply 1 , 3. This is an example of a sequence. And another example of a sequence would be, let us say, this sequence where xn is cosine of pi by 2n , sine of pi by 2n.

This sequence also each element here lies in R2 and this is similar to the previous sequence, where which also like each element like. But the sequence you can see is going to be very different. Let us just try to do that. So, what if what is x1? x1 is simply cosine of pi by 2 , sine of pi by 2, which is 0 , 1. What about x2? x2 is cosine pi , sine pi, which is this, what about x3 is going to be this, x4 is going to be this. And x5 is again going to be exactly =x1 because of the periodicity of cosine and sine.

Now, this is also a valid sequence. But you can see that this sequence does not seem to converge towards any particular point. So, I will make the notion of convergence clear soon. But you can see the difference between this sequence and this sequence. So, the first sequence as the N essentially increases, you effectively, you have your xn be a constant, but in the second sequence, as n increases, it does not seem to stay still it keeps on moving.

So, this gives us the first important definition of a sequence, which is convergence, or limit can say that, let us say you have a sequence x1 x2 dot …, where xi are d dimensional vectors, we can say that limit i approaches infinity, xi is =x star if there exists an epsilon > 0, if for all epsilon > 0, there exists an integer n such that your x, small n belongs to the ball centered at x star , epsilon for all n > or =capital.

Those the equivalent definition here so we can have a 2-sided arrow. When someone says the limit of the sequence is x star, what they mean technically is the statement, we will try to parse the statement quickly. What does this say this says that for whatever small epsilon that you choose, there exists some number integer n, such that, after that integer, you are xn is going to stay within a ball of radius epsilon centered at x star.

Let us try to check that for the first sequence that we had, whatever epsilon equals, which means that how small a ball that you can choose around x star, you can see that after some large

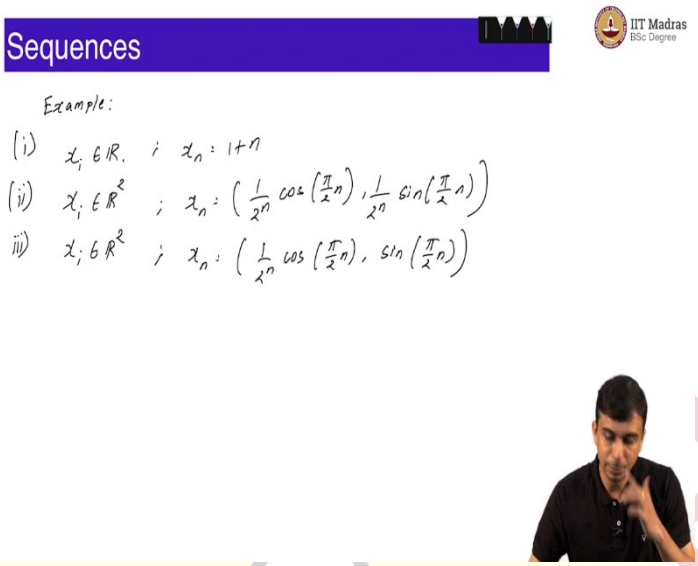
enough n, your xn's are going to stay inside that ball, you can choose however small radius that you want, you can choose a radius of 0.1, 0.01 0.001 can use whatever radius around x star which is in this case, 1 , 3, you can consider a circle or a ball of radius epsilon around 1 , 3.

And eventually, your sequence is going to go inside that ball and stay inside and it is not going to go out the polygon for, this is going to be true for whatever radius that you choose as long as it > 0. So, that this means that this particular sequence here converges to 1 , 3, or in other words, limit i approaching infinity x i is =1 , 3. So, that is the, this is an example of a convergent sequence.

You cannot say the same for the second sequence. For example, I can draw a ball of radius let us say 0.1 around any point that we can think of, let us say I draw a ball of radius 0.1 around this point. Yes, occasionally you do go your sequence does go inside this ball, but it also goes out it never stays completely inside the ball after a point. So, this particular sequence the second sequence does not converge to any point at all. So, this is an example of a non-convergent sequence.

The idea of a limit, and this also explains the logic. So, this is for all epsilon, which means that however small radius that you choose, there exists a point beyond which your sequence stays inside that ball of radius epsilon centered at x star. This is equivalent to saying that the limit i approaching infinity, xi is =x star or you might also write this as simply xi approaches x star, this is another alternate notation for writing this where your xi tends to star x star or xi approaches x star limit i is approaching infinity, x is =x dot all these are equivalent statements for the same mathematical fact.

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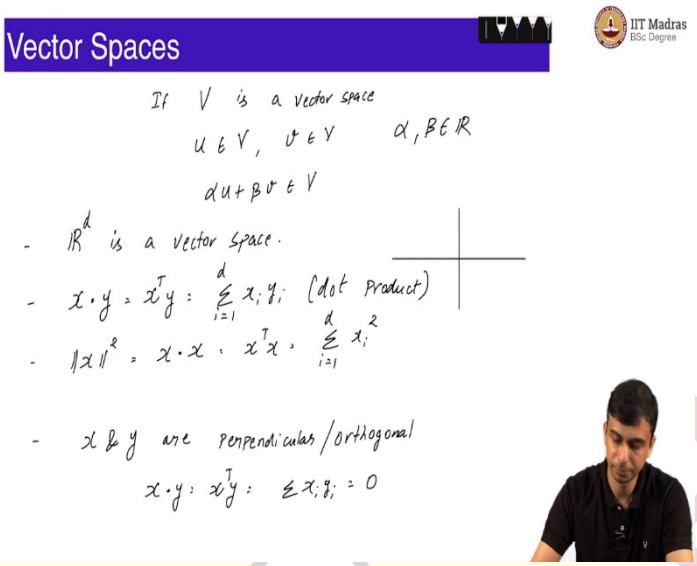


So, this here are a couple of exercise problems for you to think about. Here is one. So, for each of these examples, sequences, think, what could converge if, does the sequence converge or not? If it does converge, what is the convergent point? So, let us say we are first sequences, xi is just real value, not even R2 just real value. What is xn? xn is simply =let us say 1 + n. So, does this sequence have a convergent point? Does it converge to anything? That is the first question.

The second question is, let us say your xi are 2 dimensional vectors xi in R2. And let us say your xn is given by 1 by 2 power n, cosine of pi by 2 n , 1 by 2 power n, sine of pi by 2. The sequence looks similar to the non-convergent sequence we saw earlier, but just that now you have a1 by 2 power n and on top of it. Does this sequence converge to anything?

Then finally, you have a third sequence xi R2 where your xn is simply given by 1 by 2 power n, cosine of pi by 2 n. But the second coordinate is different. It is just sine of pi by 2 n. Just think about these 3 sequences. And do these 3 sequences have a limit? Do they converge anything? If so, what? If not, can you show that this particular these 3 sequences once these sequences converge, do not converge to anything. This is a pretty simple exercise. I hope you can do this.

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So, now we will move on to the next fundamental building block of this course, which is a vector space. A vector space is also similar to a metric space in the sense in the sense that there is also a set with an extra structure associated with it. Vector space, you can simply call this a collection of vectors, but not any arbitrary collection of vectors, this collection of vectors must satisfy certain properties, there are several properties but the most important property is that is as follows.

If v is a vector space, then this particular property must hold there are other properties but this is the most crucial property you have, let us say an element u belongs to v. And let us say u have another element v also belongs to v let us say α , $\beta$are some arbitrary real value scalars. If u, if v is a vector space, capital V is a vector space, then any linear combination of its elements is also in the in this in the sector, that is, you have α u + β V, also belongs to the vector space. This is the most crucial requirement of a vector space. There are other requirements also, but this is the most important one.

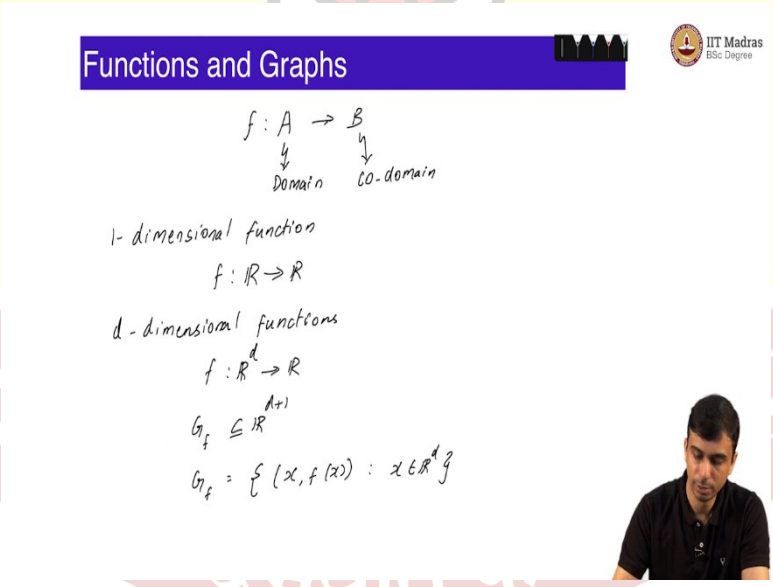
And for our purposes, the most important vector space is simply going to be Rd Rd is a vector space. Why is that just kind of obvious to see this, let us say you have 2 3 dimensional vectors. If you add them, the output is also clearly a 3-dimensional vector. Let us say you have 2 3-dimensional vectors u and v 5 ×u + 7 × V is also a 3-dimensional vector. So, Rd is clearly a vector space. According to this definition, at least it also satisfies all the other requirements of a vector space.

Some× a vector space is also associated with what is called a dot product or an inner product. You have, for example, let us say a vector, let us say x and y belongs to the vector space V, then x dot y is =x transpose y is simply sum over i =1 to d xi yi. This is the dot product of 2 vectors in the vector space and this is the dot product.

The norm of vector square norm square is simply given by x dot x or x transpose x, which is simply sum over i =1 to d x square. The vector space can also have these two properties, attributes associated with it, which is the dot product or the inner product and the norm, for our purposes we will mostly be working with Rd are subsets of it. So, it does R vector space will always have a dot product and a norm.

One notion that will be very often useful for us with vector spaces is the notion of perpendicularity or orthogonality. Let us say you have two elements x and y in a vector space, you will say x and y are perpendicular or orthogonal to each other, are orthogonal to each other, if x dot y is =x transpose y is =sigma xi yi =0. If you have the dot product of two vectors is =0, then we will call these two vectors as orthogonal or perpendicular to each other.

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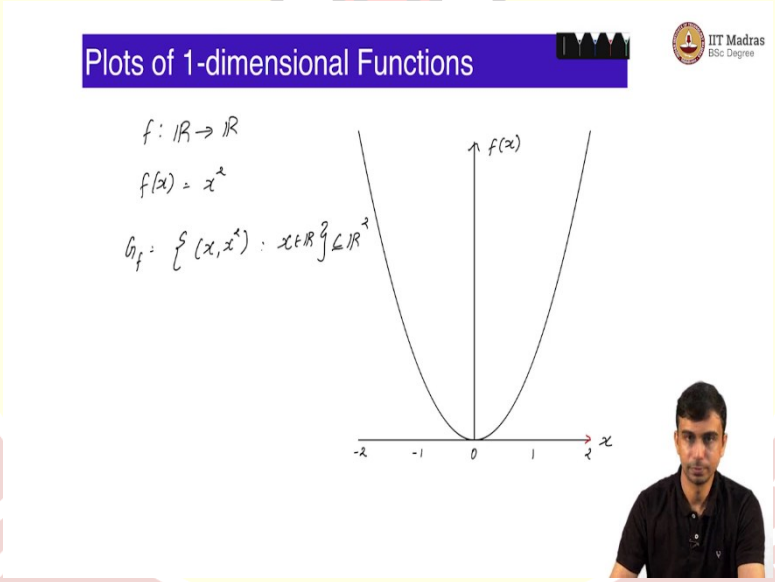


Moving on to the other important building block of our course, which is functions. Functions are simply mappings from one set to another set. So, a function f let us say is a mapping from A to B, A and B are sets is called the domain of the function f, and B is called the codomain of the function f. When we say real valued functions, we mean the coding codomain of that function is R. We will be working mostly with the real value functions.

Say one dimensional function means that you have a function from reals to reals that is, the domain is R1 and co domain is also R1. We will often also be working with D dimensional functions or multivariate functions where the function f is a mapping from Rd to R. So, that is a function a function is simply a mapping from the domain to its codomain. Sometimes for visualization purposes, it is also nice to have a graph of a function.

What is a graph of a function, a graph of a function is a subset. So, if you have a function f from Rd to R, the graph of this function, let us call it Gf, is a subset of Rd + 1. Gf is simply the set of x , f of x such that x is in Rd. The graph of a function is you vary your input over the domain, and you concatenate your input, which is an Rds, this is a D dimensional vector. This is a scalar. So, if you concatenate a d dimensional vector and a scalar, you get a d + 1 dimensional vector. So, that is why this is a subset of the graph of a function f is a subset of Rd + 1.

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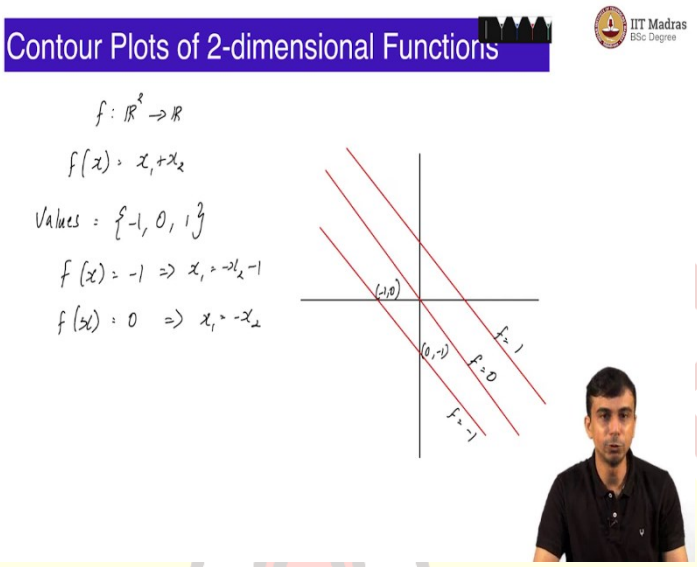


So, here are simple illustrations of visualizing simple functions first. Now, one-dimensional function, let us say one dimensional function f from R to R. The easiest way to visualize that is a simple plot where you just have an x axis, where you try different inputs and on the y axis, you simply plot f of x.

So, you for example, in this case, this function is simply f of x is =x squared. So, very simple function. And you can easily visualize this by a plot, it is possible to do that for one dimensional functions. And what is the graph of this function graph of this function is a subset of R2 now, because f is a function from R to R. The graph of this function is a subset of R2, and Gf is simply =x , x squared such that x belongs to a subset of R2. But really, it is just a, you can

imagine, the graph on a paper corresponds to that all you have ink on the paper for the scope corresponds to a subset of R2 and that is the graph of this function.

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Things become a little bit harder with two dimensional functions. That is, if you have a two-dimensional function that is f is from R2 to R, it is no longer possible to easily plot it because you will need one more dimension. And paper is just 2-dimension papers, our screens are just 2 dimension, you cannot really plot this such a function. And 2 dimensional functions are really very useful for going beyond 1D. And they are the simplest illustrations, which are non-trivial in most situations.

And so, it is nice to have a way to visualize two dimensional functions. And the standard way to visualize a two-dimensional function is to use contour plots. So, what do we mean by that, let us say we have a simple function f from R2 to R. Let us say f of x is =x1 + x2. So, x is a two-dimensional vector, which means that there has two components x1 and x2 F of x is, say x1 + x2.

How do you visualize this function, the standard would utilize this function is to think of a set of values, the output of this function can take, for example, let us say the values that you are going to give are, let us say - 1 , 0 , 1, it can be more values also, but for now, let us just say - 1 , 0 , 1. What do we mean by that, we take the function f, and we set f of x to be =- 1, we essentially can consider all solutions of this x and plot that as a curve.

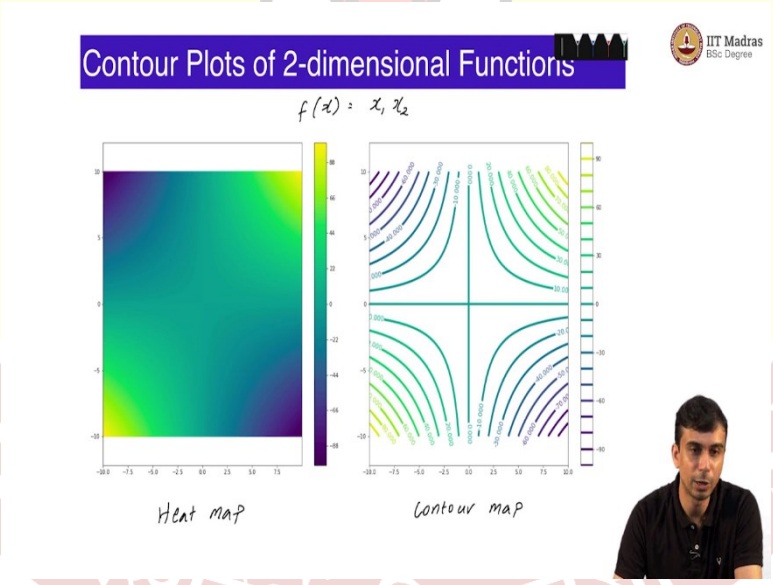
So, when you said f of x is =- 1, which means that x1 + x2 is =- 1, which x1 is =- x2 - 1. So, which means that x1, let us see, if you plot on x1 , x2 which satisfy this constraint, you would

simply get this. In particular, it would pass through the points given by 0 , - 1 - 1 coma 0. Let us label this particular red line by the contour F is =- 1.

Similarly, you can do that for other, the other values in the value set, which is f of x is =0 implies, x1 is =x2, x1 is =- x2. And that would correspond to this set. We will label this also by the property and contour which is f is =0. And similarly, you can have f =1 would correspond to this not being particularly neat inclined these figures, but this should give you an idea of what we mean by contours.

So, these for example, you could have more values here, you could have - 1, 0, 1, 2, 3, 4, 5, you would have more contacts. So, each of these lines are called contours. And this might remind you of some things you would have done in your geographies in your school. For example, you might have plotted isotherms are isobars, which are lines of constant temperature or lines of constant pressure. Those are exactly contour plots. Contour plots are the standard way to visualize 2D functions.

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Here is a contour plot of a slightly more complicated function where f of x is =x1 into x2. So, you can see that if x1 and x2 is set to 0, it means that x1 is =0 or x2 equals 0. That is why you have this particular contour labelled by 0, you can for example, set f of x is =10 and you would get this particular 4 contracts. So, you can have we can have 1 , 10 is a valid way of getting 10 - 1 or - 1 is also valid.

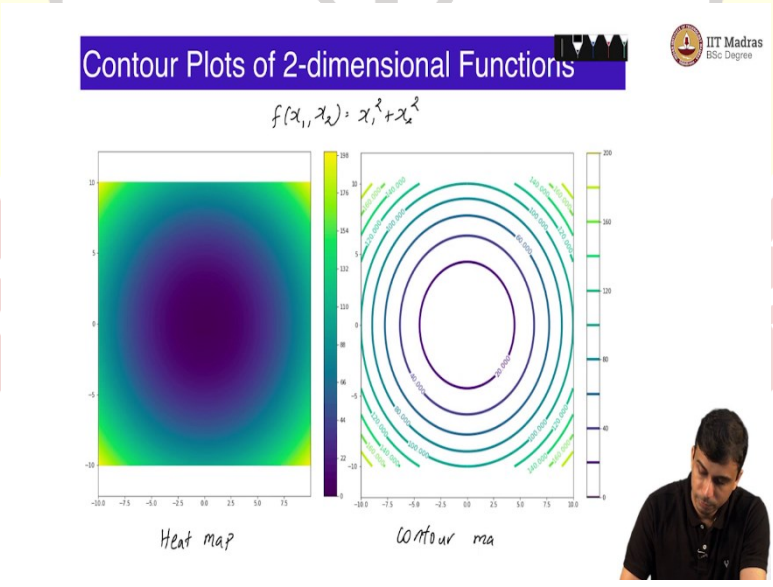
So, these both of these put together form the contour for f equals 10. Similarly, you can get f =- 10 would correspond to these two contours. This is a contour plot. And some× it is nice to

actually add color to a contour plot in technically speaking, you do not need color for a contour plot, you could just draw lines and label each contour by its value. But you could have colors also in this case, what I have done is you can see that yellow corresponds to high values, and purple corresponds to low values, and each contour has the same color.

Technically speaking, you could increase the number of contours to really large numbers instead of here, I have done I have drawn contours for 0, 10, 20, 30, 40, 50, 60, 70, 80, 90 and then - 10 - 20, - 30, - 40, and so on, but there is nothing stopping you from trying more controls for let us say 0, 1, 2, 3, 4, 5 or 0.1, 1.1, 1.2. If you drew enough contours, you would fill the entire region. If you do enough coloured contours, you would fill the entire region and that would correspond to this particular figure on the left. So, this figure is called a heat map. And the figure on the right is a contour map.

So, both have the heat map essentially gives you more information than the contour map, but some× it might be difficult to see the geometric shape of the contours from the heat map for example, here, it is not so easy to see that the shape of the contours are these curves, contour map gives that information, but the heat map technically contains more information about the function because it is infinite in many contours or there in a heat map.

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Here is one other example of another function, f of x1 , x2 is simply x1 squared + x2 squared is either contours or circles. And the heat map also looks circular, vaguely circular in shape. This is the heat map and this is a contour. So, these are two standard ways to visualize our 2D functions. With that, I think we can wrap up the basic setup and basic notation and symbols

and visualization techniques that you will need for the rest of this week. And for the rest of the course also